RAHUL KUMAR

MUNICH RELOCATION READY • EU BLUE CARD ELIGIBLE

Software Engineer — C# / .NET / Testing & Measurement Software

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PROFESSIONAL SUMMARY

Performance-driven **Software Engineer** with solid practical expertise in **C#, .NET Core, WPF (PRISM/MVVM)**, and multi-threaded system development. Proven capability in engineering reliable desktop measurement tooling, low-level serial communication interfaces (RS-232, TCP/IP sockets), and high-throughput data processing algorithms. Hands-on experience developing unit test suites (MSTest, NUnit) and CI/CD pipelines (Jenkins, Git) in cross-functional Agile/Scrum environments. English: Fluent (C1/C2), German: Elementary (A1, actively learning). Fully eligible for the **German EU Blue Card** with a recognized 4-year B.Eng in Computer Science. Dedicated to engineering robust test automation and measurement software at **ALTEN GmbH (Munich)**.

TECHNICAL EXPERTISE

Core Languages:	**C#**, **C/C++** (Fundamentals & Interop), **Python** (Scripting & Data Analysis), Java, SQL, JavaScript
.NET & Desktop Tooling:	**.NET Core / .NET 8**, **WPF** (XAML, **PRISM**, MVVM), Windows Forms, **WCF**, RESTful Web APIs, Entity Framework Core
Measurement & Systems:	**Serial Communications** (RS-232, USB, UART), **TCP/IP Socket Programming**, Hardware Interfacing, Spectrum & Vector Network Analysis (Foundations), **MATLAB** (Basic)
Quality & Testing:	**Visual Studio Unit Testing Framework (MSTest)**, **NUnit**, Moq, Automated Test Suites, TDD, Peer Code Reviews
DevOps & Tools:	**Git**, **Jenkins (CI/CD)**, **Agile / Scrum**, **Redmine**, Jira, Visual Studio 2022, Docker (Foundational)
Databases & Architecture:	**MS SQL Server**, MySQL, Stored Procedures & Indexing, Multithreading, Object-Oriented Design (SOLID)

PROFESSIONAL EXPERIENCE

Zee News • Software Developer

Aug 2024 – Present | Noida, India

Environment: C#, .NET Core, ASP.NET MVC, WCF, Serial Communication, SQL Server, MSTest, Git, Jenkins, Agile

- **Architecture & C# Engineering:** Architect and maintain high-performance C#/.NET backend services and desktop modules, optimizing streaming data pipelines and improving overall data throughput by 35%.
- **Desktop UI & Hardware Interfacing:** Engineered modular desktop user interfaces using WPF (MVVM) and Windows Forms, integrating serial communication protocols (RS-232, TCP/IP sockets) for real-time telemetry acquisition and device communication.
- **Communication Interfaces:** Developed robust Windows Communication Foundation (WCF) service interfaces and RESTful Web APIs to facilitate low-latency, asynchronous communication across distributed services.
- **Unit Testing & Quality:** Created automated unit test suites using the **Visual Studio Unit Testing Framework** and **NUnit**, raising test coverage to 85%+ and decreasing production regression defects by 40%.
- **CI/CD & Release Management:** Automated build validation and deployment pipelines using **Git** and **Jenkins**, ensuring seamless continuous integration and reliable software releases.
- **Agile Collaboration & Documentation:** Collaborated in multidisciplinary agile teams, led peer code reviews, tracked sprints in **Redmine/Jira**, and authored technical design specifications.

FEATURED ENGINEERING PROJECTS

Real-Time Telemetry & Measurement Monitoring System (ElectraSync)

C#, .NET Core, SQL Server, Multithreading, Git

- Engineered a real-time data acquisition and visualization platform handling high-frequency streaming data with sub-second dashboard updates and instant threshold alerts.
- Designed multi-threaded data ingestion routines and optimized SQL Server database indexing, reducing query latency by 45% under continuous concurrent load.
- Implemented comprehensive unit test suites ensuring zero-data-loss synchronization and fail-safe recovery.

Modular Instrument Control & Signal Visualization Tool

C#, WPF, PRISM, Python, Serial/TCP Socket Protocols

- Developed a modular measurement tooling prototype in **C#** and **WPF** using **PRISM (MVVM)** architecture to simulate instrument parameter configuration and telemetry packet parsing.
- Integrated serial communications and TCP socket handlers to ingest simulated spectrum/frequency sweep data, plotting dynamic waveform curves in real time.
- Created automated **Python** scripts for mathematical validation of signal samples against theoretical MATLAB models.

Automated Computation & Relational Data Engine

Java, MySQL, Relational Database Modeling, Algorithms

- Constructed an automated processing engine replacing manual calculations with deterministic tier-based computational algorithms, reducing processing turnaround time by 60%.
- Modeled robust relational schemas and indexed transactions in MySQL, ensuring high data integrity and consistency.

EDUCATION

Bachelor of Engineering: Computer Science & Engineering

BRCM College of Engineering & Technology

Graduated with Distinction • **GPA: 8.5 / 10.0** (Haryana, India)

Focus: Software Architecture, Algorithms, Computer Networks, Systems Design

LANGUAGES & RELOCATION

English: Fluent / Full Professional Proficiency (C1/C2)

German: Elementary / A1 (Actively learning for workplace integration)

Relocation: Available immediately for Munich; eligible for EU Blue Card